

3. News & Sentiment

The corroboration pillar. Answers: does the outside world agree with what Polymarket is pricing? A price move backed by a credible news event is far more trustworthy than one with no external anchor.

3.1 X (Twitter) Filtered Stream — WebSocket

Endpoint	Type	Auth	Purpose
https://api.twitter.com/2/tweets/search/stream	WebSocket (Server-Sent Events)	Bearer token (OAuth 2.0)	Real-time tweets matching keywords → immediate sentiment scoring

Set up filter rules per market using the Rules endpoint before connecting the stream:

```
POST https://api.twitter.com/2/tweets/search/stream/rules
Body: { "add": [{ "value": "candidate name OR event keyword", "tag": "market_id" }] }
```

Each tweet arriving on the stream is passed immediately to FinBERT for sentiment scoring. The output feeds the corroboration pillar in real time.

Signal to watch: Volume spike in tweets + sentiment shift in one direction + coinciding Polymarket price move = high-confidence corroboration.

3.2 News Headlines — REST

Source	Endpoint	Key parameters	Purpose
NewsAPI	https://newsapi.org/v2/everything	q=<keywords>, sortBy=publishedAt, language=en	Recent headlines from major outlets — Bloomberg, Reuters, NYT, etc.
GDELT	https://api.gdeltproject.org/api/v2/doc/doc	query=<keywords>, mode=artist, format=json	Free, high-volume news index. Good for volume/sentiment trend, lower quality than NewsAPI

Polling frequency: Every 2–3 minutes per active market. Most news content is neutral — focus on detecting spikes, not baseline sentiment levels.

High-confidence signal: Sentiment spike in Bloomberg/Reuters + Polymarket price jump + VPIN spike all occurring within the same short window. If the price moves without any of these anchors, treat it as uninformative flow.

3.3 FinBERT — Local NLP Model

Component	Type	Purpose
ProsusAI/finbert (HuggingFace)	Local inference (Python)	Classify each headline/tweet as positive, negative, or neutral for a financial context
Sentiment aggregator	Custom logic	Combine per-article scores into a rolling market-level sentiment signal

```
from transformers import pipeline
sentiment = pipeline("text-classification", model="ProsusAI/finbert")
result = sentiment("Candidate X wins key endorsement")
# Output: [{'label': 'positive', 'score': 0.94}]
```

Run FinBERT locally — do not call a remote API for this. Latency matters. Each headline or tweet should be scored within milliseconds of arrival.

⚠ FinBERT is trained on financial news, not prediction market text specifically. It works well for political/economic events but may misclassify niche sports or entertainment markets. Consider fine-tuning on Polymarket-specific data later.